

TECHNICAL DOCUMENTATION

AI-Powered Lead Management & Sales Automation Platform

Project Information	Details
Project Title	AI-Powered Lead Management & Sales Automation Platform
Assigned Topic	AI Sales Automation System — From Lead Capture to CRM, Qualification & Follow-up
Primary Automation Platform	n8n
Main Storage Layer	Airtable / CRM (current implementation)
Deployment	n8n Cloud / Docker, according to the deployed workflow
Documentation Purpose	Explain the complete system in beginner-friendly language

Table of Contents

1. Project Overview
2. Problem Statement
3. Project Objectives
4. What the System Does
5. System Architecture
6. Technologies and Components
7. Workflow 1 — AI Sales Automation System
8. Workflow 2 — User Lead Query Agent

9. Workflow 3 — Error Handling and Notifications

10. CRM / Database Structure

11. AI Lead Qualification and Scoring

12. Hot, Warm and Cold Lead Routing

13. Personalized Email Automation

14. Testing and Expected Results

15. Deployment

16. Security and Credential Handling

17. GitHub Repository

18. Demo Video

19. Troubleshooting Guide

20. Future Improvements

21. Conclusion

1. Project Overview

The AI-Powered Lead Management & Sales Automation Platform is an automated system built with n8n. Its purpose is to reduce repetitive sales work by taking a new lead from the moment the lead submits a form through data cleaning, duplicate checking, storage, AI qualification, lead routing, personalized email communication and CRM updates.

The project also contains a separate AI Lead Query Agent. This agent allows a user to ask questions about a lead and retrieve the appropriate information from the CRM. A separate error-handling workflow records failures and sends notifications so that problems can be noticed quickly.

Beginner explanation

Think of the system as a digital sales assistant. Instead of a salesperson manually checking every new lead, the automation receives the lead, checks it, asks AI to evaluate it, sends the appropriate email, and updates the CRM automatically.

2. Problem Statement

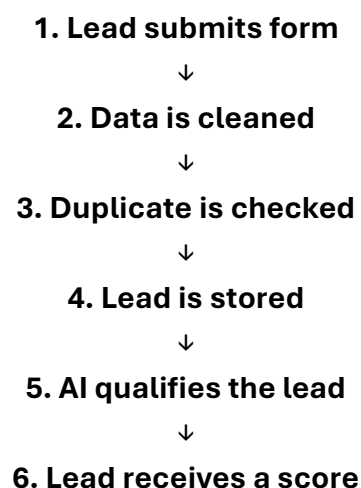
When leads are handled manually, sales teams may spend time copying data between forms and CRM systems, checking whether a lead already exists, deciding which leads are important, writing repetitive emails and updating lead status. Manual processing can also cause duplicate records, inconsistent data and missed follow-ups.

This project solves these problems by connecting the steps into one automated workflow.

3. Project Objectives

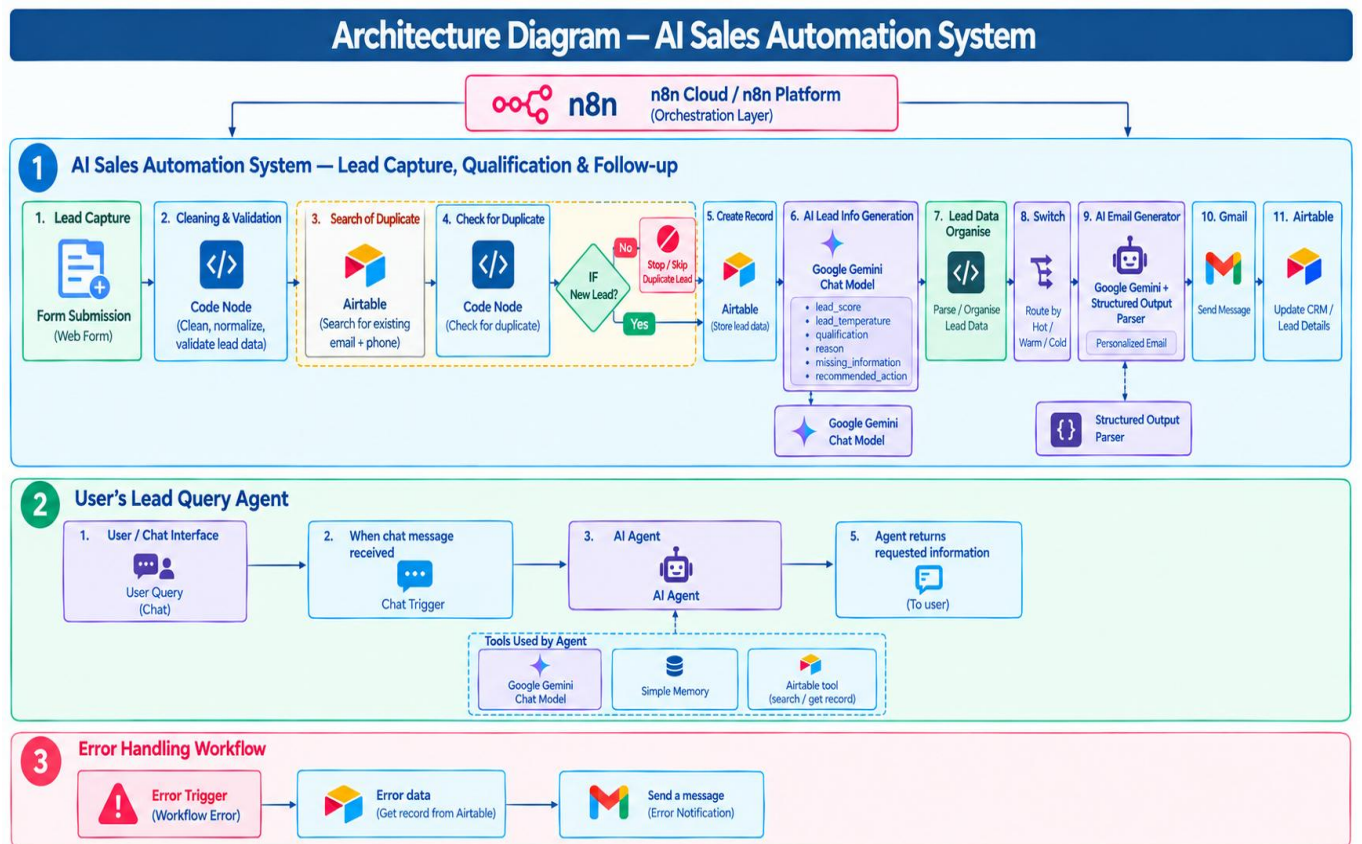
- Capture leads through a form or other supported input.
- Clean and normalize incoming lead information.
- Validate important lead fields before processing.
- Check existing records and prevent duplicate lead processing.
- Store lead information in a CRM/data store.
- Use an AI/LLM to qualify leads and assign a lead score.
- Classify leads as Hot, Warm or Cold.
- Generate personalized sales emails automatically.
- Update the CRM with qualification and follow-up information.
- Provide an AI Agent for querying lead information.
- Handle workflow errors and notify the responsible user.
- Deploy the workflow using n8n Cloud and/or Docker.

4. What the System Does



- ↓
- 7. Lead becomes Hot / Warm / Cold**
- ↓
- 8. Personalized email is generated**
- ↓
- 9. CRM is updated**

5. System Architecture



6. Technologies and Components

Technology / Component	Purpose in the project
n8n	Connects all steps and controls the automation flow.
Form / Input	Captures new lead information.
Code node	Cleans, normalizes and transforms data.

CRM / Airtable	Stores lead records and qualification information in the current implementation.
AI / LLM	Qualifies leads, generates scores and creates personalized email content.
Gmail / Email service	Sends personalized communication to leads.
AI Agent	Allows users to query lead information conversationally.
Webhooks / APIs	Can be used for system-to-system communication where required.
Docker / n8n Cloud	Provides the deployment environment.
GitHub	Stores exported workflow JSON and project documentation.

7. Workflow 1 — AI Sales Automation System

This is the main workflow. It processes a lead from form submission to CRM update and automated email.

7.1 Lead Capture

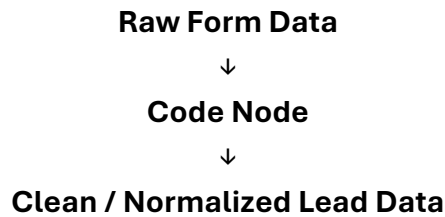
The lead starts by submitting the configured form. Typical fields in this project include Full Name, Email Address, Phone Number, Company Name, Job Title/Role, Company Size, Service Required, Estimated Budget, Timeline and submission time.

Why this step matters

The automation needs structured information before it can make decisions. A form provides consistent fields instead of relying on free-form text.

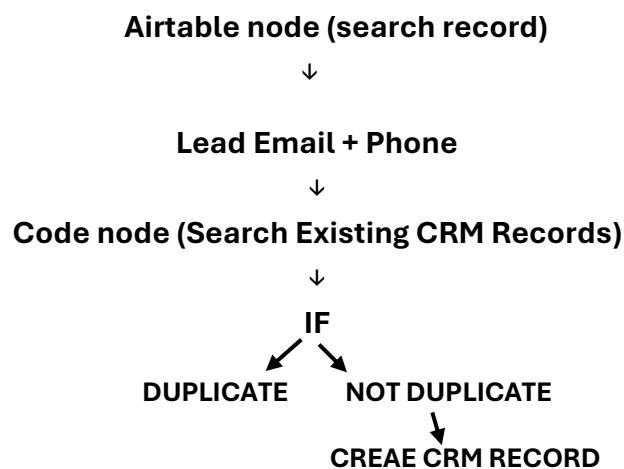
7.2 Data Cleaning and Validation

A Code node receives the raw form data and prepares it for the rest of the workflow. Typical operations include trimming unnecessary spaces, normalizing values and ensuring that important fields are available in predictable formats.



7.3 Duplicate Prevention

Before creating another lead record, the workflow checks whether the lead already exists. The project uses email and phone information as the main identifying values for duplicate detection.



The exact duplicate logic should follow the final version of the n8n workflow. The important goal is to avoid processing the same lead as a completely new lead.

7.4 Store the Lead

After the lead passes the duplicate check, its information is stored in the CRM/data store. In the current implementation, Airtable is used as the lead storage layer. If PostgreSQL is added as the final database for submission, this section should be updated to describe the PostgreSQL tables and queries actually used.

7.5 Send the Lead to AI

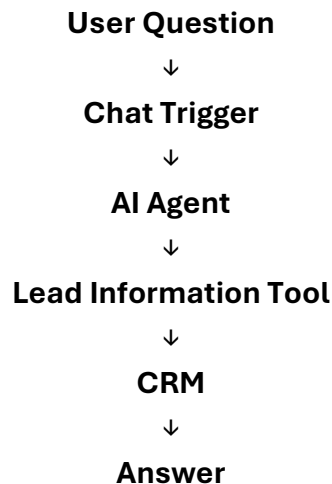
The lead information is passed to the AI qualification step. The AI evaluates the available information and returns structured qualification data rather than an unstructured paragraph.

7.6 Creating personalized email.

After read the leads and craft the message for them then it send mail to the user and update data in airtable CRM.

8. Workflow 2 — User Lead Query Agent

The User Lead Query Agent is a separate workflow designed to let a user ask questions about leads without manually opening the CRM.



8.1 How it works

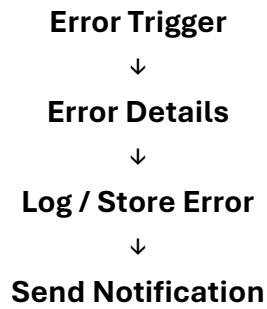
1. The user starts a conversation with the agent.
2. The agent understands the user's request and asks for identifying information such as an email when required.
3. The agent uses the lead information tool to search the CRM.
4. For lead information questions, the agent uses the appropriate stored qualification/updated information.
5. If the user asks for a different field that is not present in the updated information, the agent can check the corresponding CRM field.
6. The agent returns the result in a readable format.

Example

User: What is the lead status for this email? → Agent searches the CRM → Agent returns the relevant status/qualification information.

9. Workflow 3 — Error Handling and Notifications

Automation can fail because of invalid data, expired credentials, API errors, rate limits, unavailable services or configuration problems. The error-handling workflow is designed to capture these failures.



Useful error information can include:

- Workflow name or ID
- Failed node
- Error message
- Execution information
- Time of failure
- Retry-related information when available

The notification allows the responsible person to investigate the problem instead of discovering it much later.

10. CRM / Database Structure

The lead record should contain the information required for sales processing and AI qualification.

Field	Meaning
Full Name	Name of the lead.
Email Address / Email	Primary email used for communication and duplicate checking.
Phone Number	Contact number and secondary duplicate-check value.
Company Name	Organization associated with the lead.
Job Title	Role of the person who submitted the lead.
Company Size	Approximate size of the company.
Service Required	Service or solution requested by the lead.

Estimated Budget	Budget information provided by the lead.
Lead Score	Numeric score assigned by the AI qualification step.
Lead Temperature	Hot, Warm or Cold classification.
Status	Current processing/sales status.
Updated Information	Relevant AI-generated qualification information used by the query agent.

Database requirement

The project brief lists PostgreSQL/MySQL/Airtable/Sheet as possible storage options, while the deliverables also mention PostgreSQL. If the final submission uses Airtable, describe it honestly as the current storage implementation. Do not claim PostgreSQL was used unless it is actually connected and tested.

11. AI Lead Qualification and Scoring

The AI qualification step converts lead information into structured sales information. A typical response in this project contains the following fields:

Output	Purpose
lead_score	Numeric estimate of how valuable/ready the lead is.
lead_temperature	Hot, Warm or Cold classification.
qualification	Whether the lead appears qualified based on the available information.
reason	Short explanation for the score/classification.
timeline	Lead's stated or processed timeline when available.

Important

The AI score should not be treated as an absolute truth. It is an automated sales-prioritization signal based on the information provided to the model.

12. Hot, Warm and Cold Lead Routing

After the AI qualification step, the workflow routes the lead according to the classification returned by the AI.

Category	Meaning	Typical action
Hot	High-priority lead with strong buying signals.	Fast sales follow-up / priority contact.
Warm	Potential lead but more information or nurturing may be needed.	Follow-up and additional qualification.
Cold	Lower immediate buying intent or insufficient buying signals.	Nurturing / lower-priority follow-up.

The exact score thresholds are defined by the final workflow/prompt and should be documented here if they are explicitly configured in the submitted workflow.

13. Personalized Email Automation

The email step uses lead information and AI-generated qualification context to create a personalized message.

Lead + Qualification



AI Email Generator



Personalized Email



Gmail / Email Service



Lead

A good automated email should:

- Use the lead's name when appropriate.
- Reflect the service or requirement submitted by the lead.
- Match the lead's qualification/temperature.
- Use a clear and short call to action.
- Avoid exposing internal AI scoring details unless intentionally designed to do so.

14. Testing and Expected Results

Testing confirms that each important path works before the project is submitted.

Test Case	Expected Result	Status to record
New lead submission	Lead enters the workflow and is processed.	Pass
Invalid or incomplete data	Validation/cleaning handles the data or stops it safely.	Pass
Duplicate email + phone	Existing lead is detected and duplicate processing is prevented.	Pass
High-quality lead	AI produces a high score / Hot classification when appropriate.	Pass
Medium-quality lead	AI produces Warm classification when appropriate.	Pass
Low-quality lead	AI produces Cold classification when appropriate.	Pass
Email generation	Personalized email is generated successfully.	Pass
CRM update	Score, temperature/status and required information are updated.	Pass

Lead Query Agent	Agent finds and returns the requested lead information.	Pass
Workflow error	Error workflow captures the failure and sends notification.	Pass

Evidence

For the final PDF, add screenshots next to the most important tests: a successful lead run, a duplicate run, an AI qualification result, the CRM record, the query agent response and the error notification.

15. Deployment

The completed workflow can be deployed on n8n Cloud or a self-hosted n8n instance running in Docker. The final documentation should state which environment was used for the submitted demonstration.

15.1 n8n Cloud

- Open the deployed n8n workspace.
- Import the exported workflow JSON if needed.
- Configure credentials using n8n's credential system.
- Test the workflow manually.
- Activate the workflow after successful testing.

15.2 Docker deployment

- Run the n8n container with the required environment configuration.
- Expose the required n8n port.
- Persist the n8n data volume so workflows and configuration are not lost.
- Configure the required credentials and integrations.
- Test the production workflow before activation.

Do not document a deployment you did not actually use

Replace this section with the exact deployment setup used for the final project demonstration.

16. Security and Credential Handling

- Do not store API keys, passwords, OAuth tokens or access tokens inside the GitHub repository.
- Use n8n Credentials for connected services whenever possible.
- Do not paste private credentials into Code nodes or prompts.
- Use sample/demo lead data in screenshots when possible.
- Review exported workflow JSON before uploading it to a public repository.
- Restrict access to the CRM and Google Drive submission according to the faculty's required permissions.

GitHub warning

Exported n8n JSON can contain configuration details. Always inspect the file before making a repository public. If a secret is accidentally exposed, revoke/rotate it immediately.

19. Troubleshooting Guide

Problem	Possible cause	What to check
No data reaches the next node	Previous node returned no items or condition did not match.	Inspect the previous node's output and IF/filter conditions.
Duplicate check behaves incorrectly	Email/phone comparison or search condition is wrong.	Inspect the exact values and CRM search/filter logic.
AI returns invalid JSON	Prompt/model response is not structured as expected.	Require strict JSON output and validate/parse it before routing.
Email does not send	Credential, OAuth or provider issue.	Check the email credential and node error details.
API rate limit/quota error	Too many requests or insufficient quota.	Check provider limits, retry strategy and usage.
AI Agent cannot find a lead	Search field or CRM tool configuration is incorrect.	Check the email/identifier passed to the CRM tool.

Error notification does not arrive	Error workflow or email credential is misconfigured.	Run a controlled test error and inspect the Error Trigger output.
Workflow works manually but not automatically	Trigger/activation or deployment issue.	Check trigger configuration, activation status and deployment environment.

20. Future Improvements

- Add WhatsApp or SMS follow-ups.
- Add automatic meeting scheduling.
- Create a sales dashboard showing lead volume and conversion trends.
- Add scheduled lead-nurturing sequences.
- Add more CRM integrations.
- Add stronger validation and retry logic for external APIs.
- Add role-based access for lead information.
- Use multiple specialized AI agents for qualification, research and follow-up.
- Add monitoring and execution analytics.

21. Conclusion

The AI-Powered Lead Management & Sales Automation Platform demonstrates how n8n can be used to connect lead capture, data processing, CRM storage, AI qualification, sales routing, email communication, CRM updates and conversational lead querying into one automated system.

The project reduces repetitive manual work and gives a sales team a consistent process for handling incoming leads. The addition of duplicate prevention, error handling and an AI Lead Query Agent makes the solution more complete than a simple email automation.

The final submission should contain the working n8n workflow exports, architecture diagram, technical documentation, GitHub repository and a short demonstration video, with the required project files and blog submitted through the two Google Drive links specified by the faculty.

